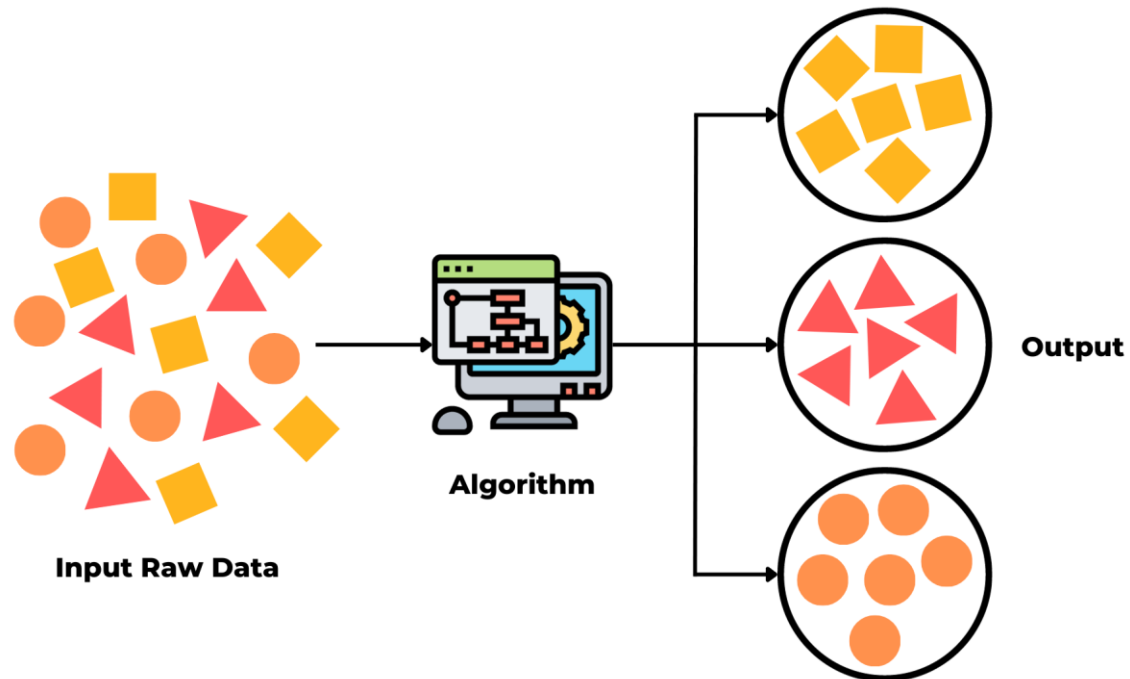


Unsupervised Learning in AIoT

Unsupervised learning identifies **patterns and structures** in data **without predefined labels**.



Why Unsupervised Learning?

- ✓ Large volumes of sensor data
- ✓ Lack of labeled datasets
- ✓ Dynamic operating environments
- ✓ Unknown failure patterns
- Manual labeling is costly and impractical.

Unsupervised Learning

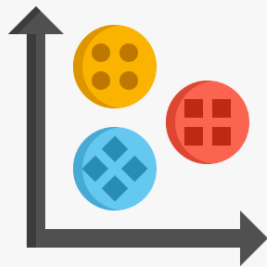
- Learns from unlabelled data
- Discovers hidden patterns
- Detects unknown behaviours

Unsupervised Learning in AIoT

AIoT Systems Use It To



**Anomaly
detection**



**Clustering
device
behaviour**



**Identifying
unknown
operating
states**

Common AIoT Applications



Equipment behaviour clustering



Energy usage pattern discovery



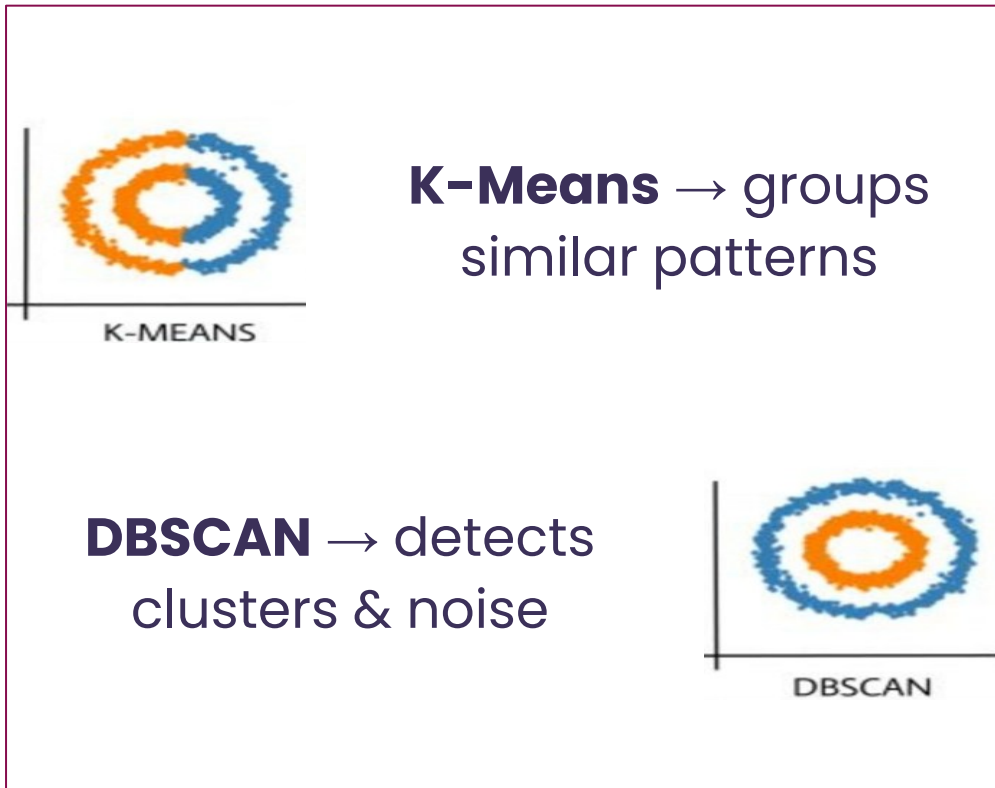
Network traffic pattern analysis



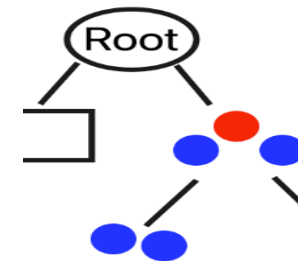
**Fleet & vehicle usage
segmentation**

Common Algorithms

Clustering Algorithms



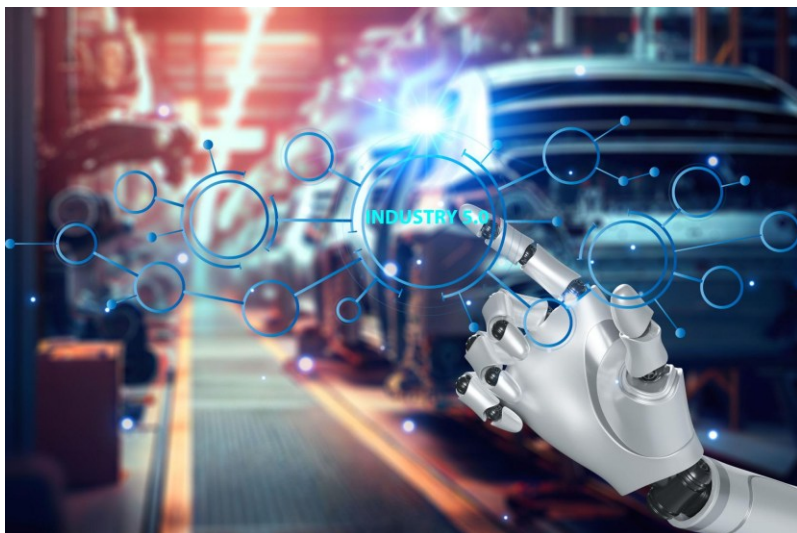
Anomaly Detection



Isolation Forest → isolates abnormal patterns

Real-Life Use Case:

Detecting Unknown Equipment Behavior in a Smart Factory



Scenario

- A manufacturing plant operates hundreds of motors and rotating machines. Each machine continuously sends:
 - vibration data
 - temperature readings
 - energy consumption
 - operating speed
- However, **there is no labeled failure data** for many machines.

**Unexpected
equipment failure
causes:**

✗ production
stoppage

✗ expensive
downtime

✗ safety risks

✗ emergency
repair costs

Traditional monitoring only detects failures **after they occur**.

Real-Life Use Case:

Detecting Unknown Equipment Behavior in a Smart Factory

Solution Using Unsupervised Learning

- Engineers apply clustering and anomaly detection:
 - ✓ K-Means groups normal operating patterns
 - ✓ DBSCAN identifies abnormal behavior clusters
 - ✓ Isolation Forest flags unusual sensor patterns
- The system learns **normal behavior patterns automatically**.

What the System Discovers

- One motor shows a **new vibration pattern** not seen before.
- Temperature increases slightly during specific load cycles.
- Pattern does not match known operating states.
- Maintenance team inspects the machine.

Outcome

- ✓ Early bearing wear detected
- ✓ Failure prevented before breakdown
- ✓ Downtimes avoided
- ✓ Maintenance scheduled proactively

Real-Life Use Case:

Detecting Unknown Equipment Behavior in a Smart Factory

Key Learning Takeaways

- ✓ Unsupervised learning works without labeled data
- ✓ It discovers unknown problems
- ✓ It prevents failures before they occur
- ✓ It enables predictive maintenance

Discussion Question for Learners

Why might this issue not be detected using rule-based monitoring?